```
    input1 = input("What is your username: ")
while input2 != password:
    input2 = input("What is your password: ")
print("Welcome back to your system!")
```

If you would like the program to run continuously, just add a `while 1 == 1:` loop around the whole thing. You will have to indent the rest of the program when you add this at the top of the code, but don't worry, you don't have to do it manually for each line! Just highlight everything you want to indent and click on "Indent" under "Format" in the top bar of the python window.

Another way of doing this could be:

```
name = input('Set name: ')
password = input('Set password: ')
while 1 == 1:
    nameguess=""
    passwordguess=""
    key=""
    while (nameguess != name) or (passwordguess != password):
        nameguess = input('Name? ')
        passwordguess = input('Password? ')
    print("Welcome,", name, ". Type lock to lock.")
    while key != "lock":
        key = input("")
```

Notice the `or` in while `(nameguess != name) or (passwordguess != password)`, which we haven't yet introduced. You can probably figure out how it works.

ca:Python 3 per a no programadors/Comptar fins a 10[1]

1 https://ca.wikibooks.org/wiki/Python%203%20per%20a%20no%20programadors%2FComptar%20fins%20a%2010

6 Decisions

6.0.1 If statement

As always, I believe I should start each chapter with a warm-up typing exercise, so here is a short program to compute the absolute value of an integer:

```
n = int(input("Number? "))
if n < 0:
   print("The absolute value of", n, "is", -n)
else:
   print("The absolute value of", n, "is", n)
```

Here is the output from the two times that I ran this program:

```
Number? -34
The absolute value of -34 is 34
```

```
Number? 1
The absolute value of 1 is 1
```

So what does the computer do when it sees this piece of code? First it prompts the user for a number with the statement ”`n = int(input("Number? "))`”. Next it reads the line ”`if n < 0:`”. If `n` is less than zero Python runs the line ”`print("The absolute value of", n, "is", -n)`”. Otherwise it runs the line ”`print("The absolute value of", n, "is", n)`”.

More formally Python looks at whether the *expression* `n < 0` is true or false. An `if` statement is followed by an indented *block* of statements that are run when the expression is true. Optionally after the `if` statement is an `else` statement and another indented *block* of statements. This second block of statements is run if the expression is false.

There are a number of different tests that an expression can have. Here is a table of all of them:

operator	**function**
`<`	less than
`<=`	less than or equal to
`>`	greater than
`>=`	greater than or equal to
`==`	equal
`!=`	not equal

Another feature of the `if`command is the `elif`statement. It stands for else if and means if the original `if`statement is false but the `elif`part is true, then do the `elif` part. And if neither the `if` or `elif` expressions are true, then do what's in the `else` block. Here's an example:

```
a = 0
while a < 10:
    a = a + 1
    if a > 5:
        print(a, ">", 5)
    elif a <= 3:
        print(a, "<=", 3)
    else:
        print("Neither test was true")
```

and the output:

```
1 <= 3
2 <= 3
3 <= 3
Neither test was true
Neither test was true
6 > 5
7 > 5
8 > 5
9 > 5
10 > 5
```

Notice how the `elif a <= 3` is only tested when the `if` statement fails to be true. There can be more than one `elif` expression, allowing multiple tests to be done in a single `if` statement.

6.0.2 Examples

```
# This Program Demonstrates the use of the == operator
# using numbers
print(5 == 6)
# Using variables
x = 5
y = 8
print(x == y)
```

And the output

```
False
False
```

high_low.py

```
# Plays the guessing game higher or lower

# This should actually be something that is semi random like the
# last digits of the time or something else, but that will have to
# wait till a later chapter.  (Extra Credit, modify it to be random
# after the Modules chapter)
number = 7
guess = -1

print("Guess the number!")
while guess != number:
```

```
    guess = int(input("Is it... "))

    if guess == number:
        print("Hooray! You guessed it right!")
    elif guess < number:
        print("It's bigger...")
    elif guess > number:
        print("It's not so big.")
```

Sample run:

```
Guess the number!
Is it... 2
It's bigger...
Is it... 5
It's bigger...
Is it... 10
It's not so big.
Is it... 7
Hooray! You guessed it right!
```

even.py

```
# Asks for a number.
# Prints if it is even or odd

number = float(input("Tell me a number: "))
if number % 2 == 0:
    print(int(number), "is even.")
elif number % 2 == 1:
    print(int(number), "is odd.")
else:
    print(number, "is very strange.")
```

Sample runs:

```
Tell me a number: 3
3 is odd.
```

```
Tell me a number: 2
2 is even.
```

```
Tell me a number: 3.4895
3.4895 is very strange.
```

average1.py

```
# keeps asking for numbers until 0 is entered.
# Prints the average value.

count = 0
sum = 0.0
number = 1 # set to something that will not exit the while loop immediately.

print("Enter 0 to exit the loop")

while number != 0:
```

```
    number = float(input("Enter a number: "))
    if number != 0:
        count = count + 1
        sum = sum + number
    if number == 0:
        print("The average was:", sum / count)
```

Sample runs

Sample runs:

```
Enter 0 to exit the loop
Enter a number: 3
Enter a number: 5
Enter a number: 0
The average was: 4.0
```

```
Enter 0 to exit the loop
Enter a number: 1
Enter a number: 4
Enter a number: 3
Enter a number: 0
The average was: 2.66666666667
```

average2.py

```
# keeps asking for numbers until count numbers have been entered.
# Prints the average value.

#Notice that we use an integer to keep track of how many numbers,
# but floating point numbers for the input of each number
sum = 0.0

print("This program will take several numbers then average them")
count = int(input("How many numbers would you like to average: "))
current_count = 0

while current_count < count:
    current_count = current_count + 1
    print("Number", current_count)
    number = float(input("Enter a number: "))
    sum = sum + number

print("The average was:", sum / count)
```

Sample runs:

```
This program will take several numbers then average them
How many numbers would you like to average: 2
Number 1
Enter a number: 3
Number 2
Enter a number: 5
The average was: 4.0
```